MIL-C-58052C(AR)
15 March 1985
SUPERSEDING
MIL-C-58052B(MU)
18 July 1969

MILITARY SPECIFICATION

COMPASS, MAGNETIC, UNMOUNTED: M2 (NONMAINTAINABLE)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the design, performance, and inspection requirements for one type of combination compass and clinometer. No repairs or replacement of parts are anticipated for this instrument.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

DD-G-451	-	Glass, Float or Plate, Sheet, Figured
GG-P-455	-	Plates and Foils, Photographic (Photosensitive Anodized Aluminum)
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
TT-E-529	-	Enamel, Alkyd, Semi-gloss
MMM-A-130	-	Adhesive, Contact

FSC 1290

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-3144	-	Vials, Level
MIL-M-13508	-	Mirrors, Glass, Front Surfaces Aluminized for Optical Elements
MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-171	-	Finishing of Metal and Wood Surfaces

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

PACKAGING DATA SHEET

10547166	-	Packaging of Compass M2 with Carrying Case

OTHER PUBLICATIONS

MIL-HDBK-204	-	Inspection Equipment Design

(Copies of military handbooks should be obtained from the Superintendent of Documents, U.S. Government Pringint Office, Washington, D.C. 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Design. Design requirements shall conform to all requirements and figures herein to insure acceptable construction and final acceptance of a non-maintainable Compass: M2.

3.2 First article. When specified, a sample shall be subjected to First Article Inspection. (See 4. and 6.)

3.3 Construction.

3.3.1 General description. The instrument shall consist of a magnetic needle compass, a clinometer for measuring of angles in the vertical plane, a front and rear leaf sight for establishing a line of sight, and an adjustable mirror for viewing the compass and clinometer scales while sighting. A circular level vial shall be provided for indicating the level position of the compass box. An azimuth scale adjusting mechanism shall be provided to permit setting a declination constant. A needle lifting mechanism shall be incorporated to lift the compass needle from its pivot and secure it when not in use.

3.3.2 Configuration. The general configuration of the compass is shown on Figures 1 and 2.

3.3.3 Size and weight. The dimensional limits shown in Figures 1 and 2 shall not be exceeded. The weight of the compass shall not exceed 10 ounces.

3.3.4 Compass box. The compass and clinometer shall be contained in an airtight housing having a transparent cover which provides for full visibility of the azimuth scale and index, clinometer scale and index, clinometer level vial. The compass level vial shall have a minimum of 60% visibility. A slotted external adjustment shall be provided by which the azimuth scale may be rotated, with respect to a fixed index, for compass declination adjustment.

3.3.4.1 Cover and mirror. The cover shall be hinged to the "South" end of the compass box. Hinge friction shall permit setting of the cover at any angle from a fully closed to a 180° open position. The mirror shall be a rear surface type and shall be aluminized in accordance with MIL-M-13508. A durable protective backing shall be applied over the aluminized surface. The mirror shall be integral with, bonded or otherwise firmly affixed to the cover. The cover and mirror shall provide a 3/8" X 15/32" transparent area and a centerline which will serve as an alternative front sight.

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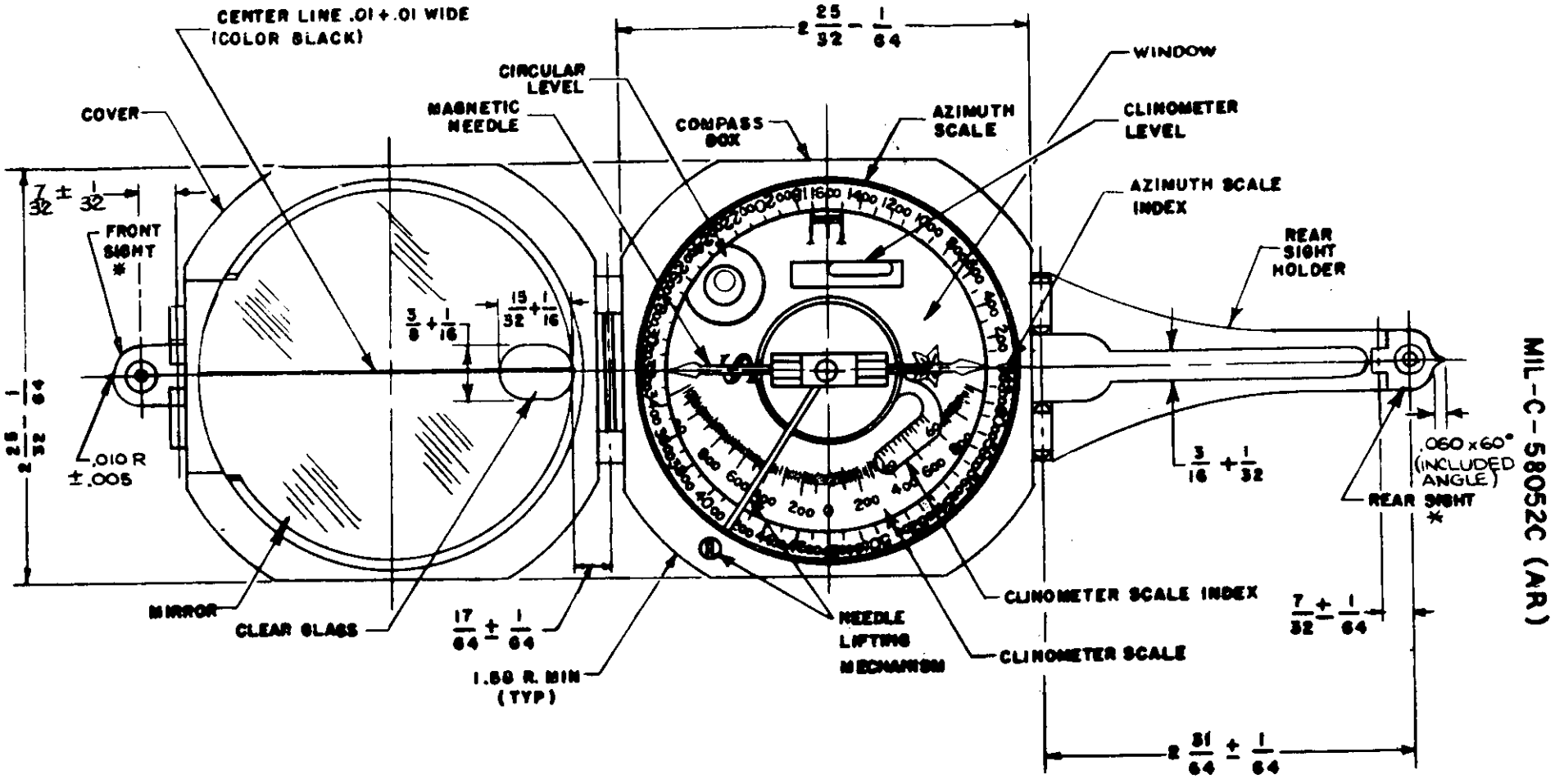
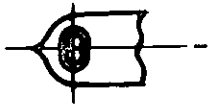
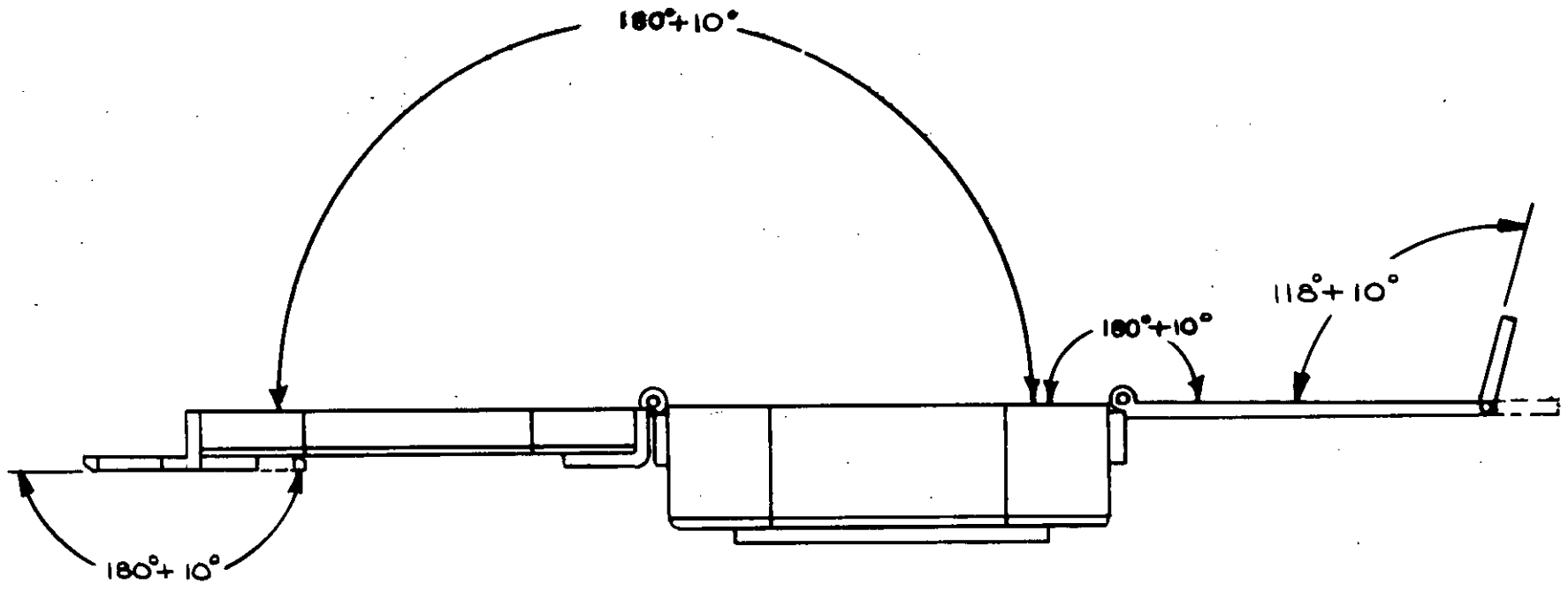


FIGURE 1. Top view, Compass: M2.

* NOTE:-
ALTERNATE DESIGN OF FIGURE EIGHT TYPE COUNTERSINK MAY BE USED.



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COMPASS OPEN SHOWING REQUIRED MOVEMENT FOR SIGHTS AND COVER

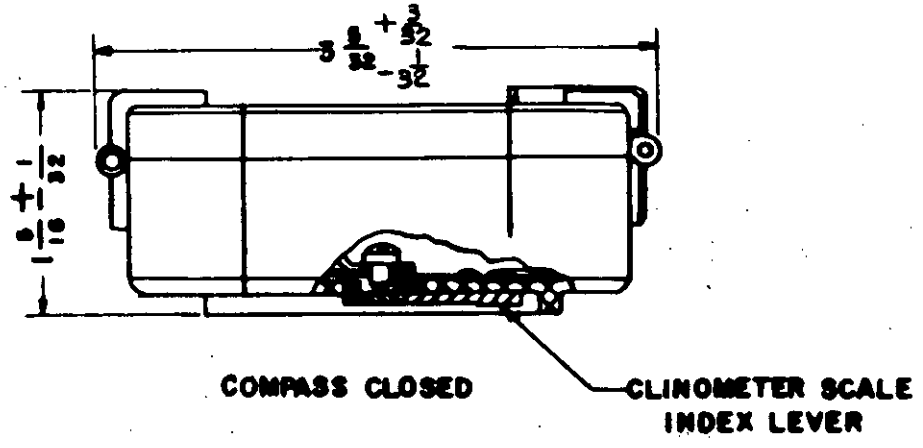


FIGURE 2. Side view, Compass: M2.

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3.3.4.2 Needle lifting mechanism. Closing the cover shall actuate a mechanism in the compass box. The mechanism shall lift the compass needle from its pivot and hold the center of the needle against the transparent cover of the compass box. The mechanism shall be capable of being actuated by light thumb pressure.

3.3.4.3 Cover fastener. The cover shall latch into the closed position by use of a spring detent or closely controlled mechanical interference. The cover shall open by use of appropriate finger grips or a push-to-open plunger-type spring release.

3.3.4.4 Compass needle configuration. The shape of the compass needle shall be symmetrical. Each end of the needle shall be suitable for azimuth readings.

3.3.4.5 Clinometer. The clinometer shall consist of a tubular level vial, clinometer scale, and index mounted in the compass box in such a manner that the level vial and index may be rotated with respect to the fixed needle by means of an external lever.

3.3.4.6 Scale markings. Scale markings shall conform to Figures 3a and 3b. The clinometer scale shall be integral with or firmly affixed to the compass box.

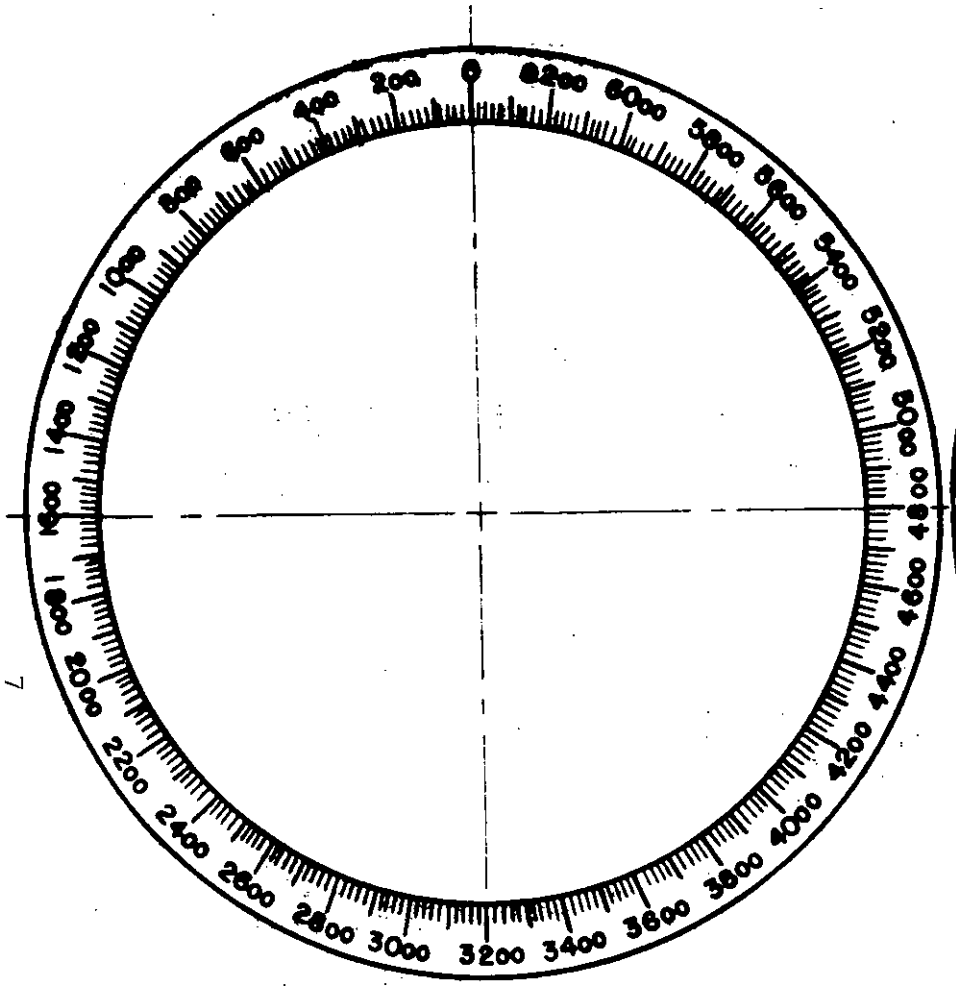
3.3.5 Sight assemblies.

3.3.5.1 Rear sight arm. The rear sight arm shall be hinged to the "North" end of the compass box (see Figures 1 and 2). Hinge friction shall permit setting of the arm at any angle between fully closed (folded over top of the compass box) and the 180 +10 degree open position. Hinge stops shall be provided at the 180 +10 degree open position.

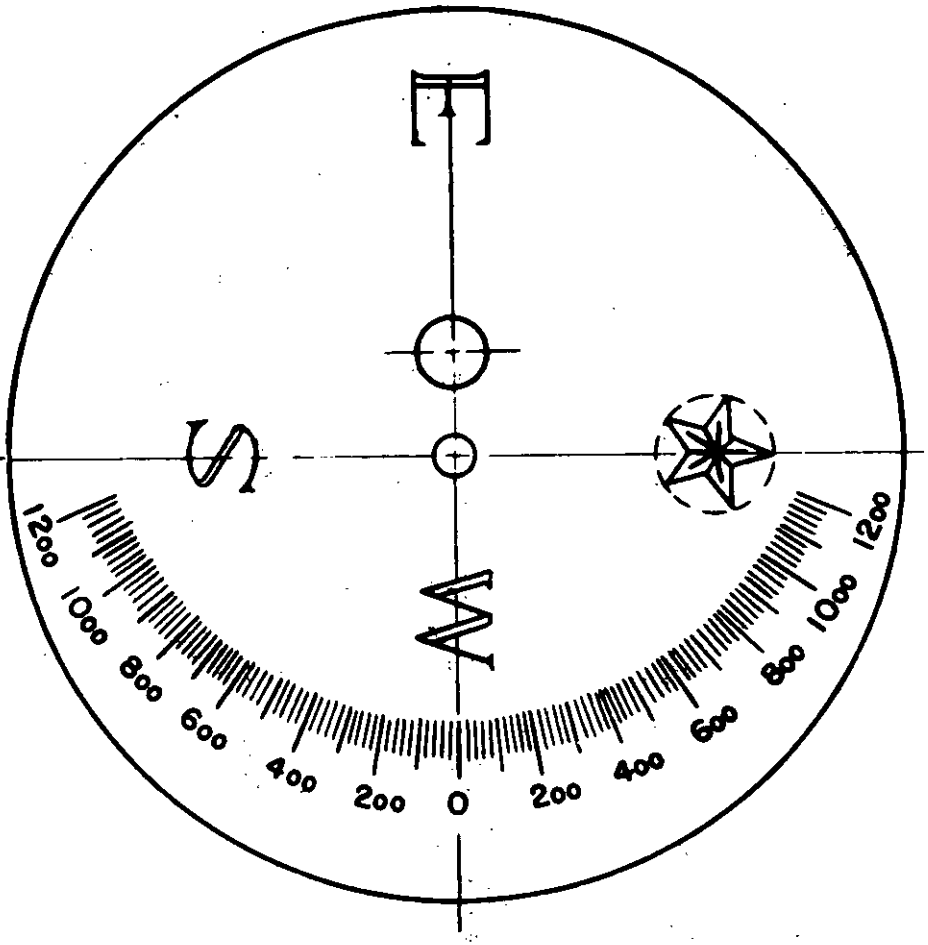
3.3.5.2 Rear sight. The rear sight shall be hinged to the rear sight arm in such a manner that the sight may be set at any angle between 118 +10 degrees and 180 +10 degrees with respect to the arm. Hinge stops shall be provided at 118 +10 and 180 +10 degrees.

3.3.5.3 Front sight. The front sight shall be hinged to the cover in such a manner as to allow setting of the sight at any angle between the full open position (at 180° with respect to the cover) to its full closed position (flat against the mirror within the cover).

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AZIMUTH SCALE
FIGURE 3a



CLINOMETER SCALE
FIGURE 3b

FIGURE 3. Scales.

NOTES:

1. GRADUATIONS TO HAVE 320 EQUAL DIVISIONS IN 360°.
2. LINES ON AZIMUTH SCALE 3/64, 5/64 AND 3/32 LONG. NUMERALS 1/32 AND 1/16 HIGH.
3. LINES ON ELEVATION SCALE 1/8, 9/64 AND 11/64 LONG. NUMERALS 1/32 AND 1/16 HIGH. LETTERS AND STAR 1/4 HIGH.
4. COLOR OF ALL LINES, NUMERALS AND STAR TO BE WHITE ON DULL OR SEMI-GLOSS BLACK BACKGROUND.

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3.3.5.4 Clearance between sights and glass. With the compass completely closed, (metal to metal contact) the inclosed front and rear sights shall have sufficient clearance between sights and glass to proclude breakage of the window and mirror due to mechanical interference.

3.3.6 Materials.

3.3.6.1 Quality. All materials shall be of uniform quality and free of defects which may impair the functioning or service life of the compass, and shall meet the requirements of DD-G-451, GG-P-455, QQ-A-250/2, TT-E-529 and MMM-A-130.

3.3.6.2 Nonmagnetic properties. Materials and finishes for all components except the compass needle and pivot shall be inherently nonmagnetic and free of magnetic impurities.

3.3.6.3 Resistance to static charge. Materials shall not build up a charge of static electricity great enough to cause improper functioning of the compass needle. The transparent compass box cover and the mirror shall not retain a charge of static electricity when rubbed with a dry cotton cloth.

3.3.6.4 Corrosion resistance. Metals shall be selected for maximum corrosion resistance. Dissimilar metals which have a tendency to produce galvanic corrosion shall not be used in intimate contact unless suitably protected against such corrosion.

3.3.6.5 Fungus resistance. Materials shall be inherently fungi static or treated with a suitable, permanent fungicial agent.

3.3.6.6 Resistance to abrasion and shatter. The compass box window shall be fabricated of a heat treated case hardened sheet glass. The mirror shall be fabricated of a heat treated case hardened sheet glass or of glass meeting the requirements of Specification DD-G-451, Type I, Class I, quality q^1, or q^2, float glass. The use of plastic material in this application is not permissible.

3.3.6.7 Light transmissibility retention. The transparent compass box cover and the 3/8" X 15/32" transparent area shall be inherently resistant to light transmission loss due to aging.

3.3.6.8 Resistance to reflectance-loss. The reflecting surface of the mirror shall be inherently free of reflectance loss due to aging or flaking.

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3.3.7 Final finishes.

3.3.7.1 Exterior. All exterior metal surfaces, (except the front and rear sights, sight hinge assemblies, latch, declination adjusting screw, clinometer lever and cover glass retaining ring), shall have a Forest Green finish equivalent to or better than requirements of finish 21.3 of MIL-STD-171.

3.3.7.2 Sights. The front and rear sights, sight hinge assemblies, latch, clinometer lever and declination adjusting screw shall have a lusterless black finish. The finish may be applied chemically, painted, impregnated or amalgamated in the material. The finish coating shall be equivalent to or better than requirements of finish 20.8 of MIL-STD-171.

3.3.7.3 Scales. The azimuth scale and the clinometer scale and markings shall be white on a lusterless or semi-gloss black background.

3.3.7.4 Compass needle. The compass needle shall be black except that the north-seeking tip shall be white.

3.3.8 Compass level vial. The compass level vial shall conform to the performance and sensitivity requirements of MIL-V-3144, Type CA1, except that a fluorescent yellow liquid shall be used. Configuration of Type CA1 or Type CA2 may be used. Type marking is not required.

3.3.9 Clinometer level vial. The clinometer level vial shall conform to the performance and configuration requirements of MIL-V-3144, Type DR1, except a fluorescent yellow liquid shall be used. Type marking is not required.

3.3.10 Identification marking. Identification conforming to data on figure 4 shall be provided on the outside of the compass cover. Nameplate size dimensions are for reference only.

3.3.10.1 Identification plate. The identification plate shall meet the applicable requirements of MIL-STD-130.

3.3.10.2 Serial numbers. A block of serial numbers, for use in identification marking, will be assigned to the contractor by the procuring agency.

3.4 Environmental.

3.4.1 Storage temperature range. There shall be no evidence of physical failure such as breakage and bonding separation or casting failure due to storage at temperatures ranging from -80° to +160° Fahrenheit (F).

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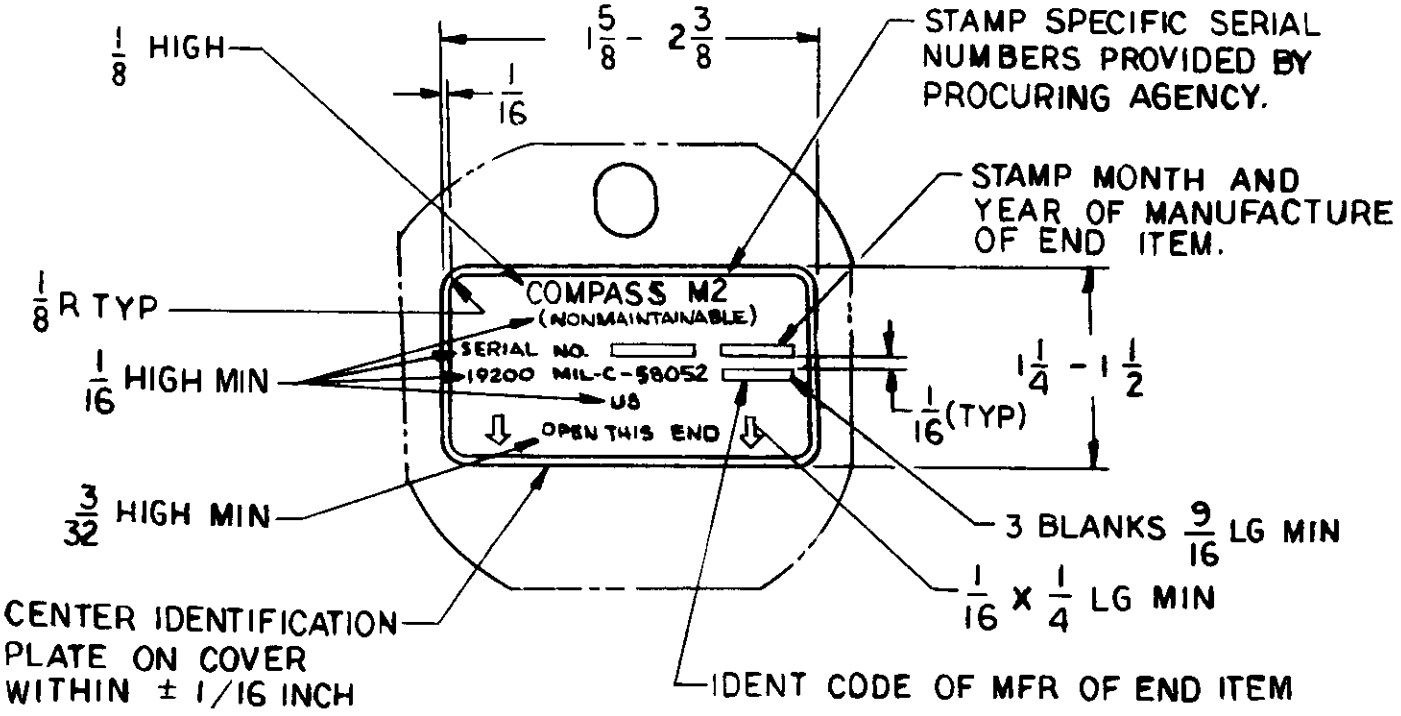


FIGURE 4. Identification plate.

BOND IDENTIFICATION PLATE TO COVER USING A PERMANENT CONTACT OR PRESSURE SENSITIVE ADHESIVE. AFTER BONDING, TOP SURFACE OF IDENTIFICATION PLATE TO BE RECESSED TO A DEPTH OF .015 MINIMUM BELOW THE TOP SURFACE OF THE COVER.

PREFERRED:

1. ALUMINUM ALLOY 3003, TEMPER H14 OR H24. SPEC QQ-A-250/2. .025 THICK
2. ALL LETTERS GOTHIC CAPITALS, NUMERALS TO BE OF SIMILAR APPEARANCE AND CENTRALLY LOCATED. LETTERS, NUMERALS, ARROWS, RIM AND BLANKS RAISED .003 TO .005, AND DULLED BEFORE ANODIZING.
3. FINISH 7.2.1 OF MIL-STD-171. BACKGROUND FILLED WITH ENAMEL, COLOR BLACK, SPEC TT-E-529.

ALTERNATE:

1. PLATE AND FOILS, PHOTOGRAPHIC, TYPE I, CLASS 1, SPEC GG-P-455, .003 THICK MIN.
2. ALL LETTERS GOTHIC CAPITALS, NUMERALS, ARROWS, RIM AND BLANKS TO BE CLEAR MATTE FINISH ON BLACK BACKGROUND.

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3.4.2 Condensation. The interior of the compass box shall be free of visible frost when the compass is thermally stabilized at -25°F.

3.4.3 Temperature conditions. Except as specified in 3.7.9 and 3.9, the following requirements shall be met at a temperature between 60° and 90°F.

3.4.4 Vibration. This requirement shall be met with the cover in the closed position. The compass shall show no evidence of physical failure due to being vibrated at 30 cycles per second and an amplitude of 0.06 inch (0.12 inch total excursion).

3.4.5 Shock. This requirement shall be met with the cover in the closed position. The compass shall show no evidence of damage due to being subjected to the equivalent of shocks resulting from two 36-inch drops onto a solid wood base covered with 4 inches of 40-grit kiln-dried sand. One drop shall be with the base of the compass horizontal; the other with the base vertical.

3.4.6 Sealing. This requirement shall be met with the cover in the open position. The compass box shall show no evidence of leakage, when submerged in water elevated to 130° to 140°F (54° to 60°C). Hold the compass submerged with the uppermost surface covered by not less than 1 inch of water for a period of at least three minutes or until a leak is detected. Rotate the compass and observe until all surfaces have been examined.

3.5 Cleanliness. The interior of the compass box shall be free of visible foreign matter.

3.6 Performance.

3.6.1 Reseatability of needle. This requirement shall be met with the cover in the fully opened position. The compass needle shall reseat itself and swing freely on its pivot subsequent to inverting and returning the compass to the use position.

3.6.2 Tilt clearance. The compass needle shall have freedom of rotation when the compass is 3° + 1° out of level and shall return to within 10 mils of a previously established position after having been deflected a minimum of 200 mils.

3.6.3 Damping. The compass needle shall come to rest within 15 seconds after being deflected 400 mils.

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3.6.4 Half-periods. The average half-period of the magnetic needle shall be 2.5 seconds or less for a minimum of six successive half-periods when the magnetic needle is deflected not more than 400 mils. The half-period is defined as the time interval between successive transits of the end of the needle through its point of magnetic equilibrium.

NOTE: This requirement applies in areas of the earth where the horizontal intensity of the magnetic field is not lower than 0.12 oersteds. Where lower intensities are encountered, comparable standards will be established by the contracting officer.

3.6.5 Declination adjustment.

3.6.5.1 Excursion range. The azimuth scale shall be adjustable through a minimum of 800 mils in each direction of the zero setting.

3.6.5.2 Backlash. Backlash in the declination adjustment movement shall not exceed 20 mils.

3.6.6 Distortion mirror. With the cover opened to an angle of 45°, no distortion shall be visible when the compass face is viewed in the mirror.

3.6.7 Reflectance. Reflectance of the mirror shall be not less than 80%.

3.6.8 Visibility, scales. This requirement shall be met with the observations made by the unaided eye. All scale markings shall be clearly defined and free of distortion when viewed directly through the transparent compass box cover and by light reflected from the mirror.

3.6.9 Visibility, target. This requirement shall be met with the observation made by the unaided eye. The image of a target shall appear clearly defined and free of distortion when viewed through the 3/8" x 15/32" transparent area.

3.7 Accuracy.

3.7.1 Compass needle horizontal alignment. This requirement shall be met with the bottom of the compass box on a horizontal plane. The top surfaces of the compass needle tips and the azimuth scale shall be on the same horizontal plane within 0.030 inch.

3.7.2 Compass needle tip clearance. Clearance between the compass needle tips and the azimuth scale shall not exceed 0.030 as measured horizontally.

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3.7.3 Alignment of sights. The requirements of 3.6.3.1 and 3.6.3.2 shall be met with the compass in the open position shown on Figure 2.

3.7.3.1 Sight alignment, azimuth. With the cover and rear sight fully open and the front sight tip at 90° and the rear sight tip at 118°, the sight tips, azimuth scale index, azimuth zero graduation, the 3200 mil graduation and the etched line on the mirror shall be in a single plane perpendicular to the base of the compass box within plus or minus .020 inch.

3.7.3.1.1 Alignment, sight tips. With the cover open to an angle of 60°, the front and rear sight tips shall be aligned within .020 inch when brought into contact with each other.

3.7.3.1.2 Alignment, rear sight tip and mirror line. The rear sight tip shall track the mirror line within .020 inch total displacement (top to bottom of line).

3.7.3.2 Sight alignment, elevation. With the compass positioned as in 3.6.3.1, the line of sight determined by the sight tips shall be parallel within plus or minus .020 inch to a plane through the west (W) side of the compass box and cover, as measured at the sight tips.

3.7.4 Compass level vial setting. This requirement shall be met with the bottom of the compass box on a horizontal plane. The bubble in the compass level vial shall be concentric with the red circle within the width of the line forming the circle. If the level vial is mounted on a moving part, this requirement shall be met at any position of the moving part.

3.7.5 Clinometer level vial setting. This requirement shall be met with the west (W) side of the compass box and cover on a horizontal plane. The clinometer scale index shall indicate zero within 10 mils when the clinometer level vial bubble is centered.

3.7.6 Clinometer readings. The compass shall be capable of meeting this requirement at any elevation setting between 1200 mils above and 1200 mils below a horizontal plane. The clinometer scale index shall indicate changes in the angle of the inclination within 10 mils when the level vial bubble is centered.

3.7.7 Azimuth readings. This requirement shall be met with the compass level vial bubble centered and the north-seeking tip of the compass needle originally indicating zero. Rotation of the compass about the geometric axis of the compass box to any measured angle shall result in a corresponding azimuth scale reading within 10 mils. The south-seeking tip shall indicate 3200 mils from any measured azimuth angle within 10 mils.

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3.7.8 Repeatability of azimuth readings. The magnetic needle shall return to within 10 mils of a previously established position after having been deflected from that position through an angle of not less that 200 mils and not greater than 800 mils at any temperature between minus 40 degrees and plus 150 degrees F.

3.7.9 Operating life. Requirements of 3.7.8 shall be met after raising and dropping the needle on the pivot 15,000 times by actuation of the needle lifting mechanism of the compass.

3.8 Operability of movable parts (60° to 90°F).

3.8.1 Effectiveness of cover fastener. This requirement shall be met with the cover originally in the closed position. Where a push-to-open latch is used, the direct force required to release the latch shall not exceed 16 ounces. The push-to-open spring latch shall provide a positive lock and shall prevent opening of the cover when a torque of 45 ounce-inch is applied. Where provided with a closely controlled mechanical interference or spring detent, the starting torque required to open the cover shall be within the limits of 16 and 48 ounce-inches above the recorded torque for the cover hinge. All types of cover fasteners shall meet the above requirements after 2000 cycles of operation.

3.8.2 Hinges and adjustments. The running torque required to position the hinged assemblies or operate the adjustable assemblies shall be within the values in Table I at standard ambient temperatures. All moving parts should be capable of maintaining any set position, and should move through its entire range without exceeding the maximum running torque values in Table I.

TABLE I. Torque requirements.

Torque values in ounce-inches

	MAX	MIN
Rear sight arm hinge	50	Should support weight of instrument
Rear sight hinge	10	Maintain position
Front sight hinge	8	Maintain position
Cover hinge	50	Should support weight of compass body
Clinometer index and level lever	9	Maintain position
Declination adjusting mechanism	8	Maintain position

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3.9 Operability of movable parts (-40° and +150°F). This requirement shall be met while the compass is thermally stabilized at -40° and +150°F. The force required to operate the push-to-open latch will not exceed 50 percent of the values listed in 3.8.1. The spring detent latch will be as listed in 3.8.1. The minimum values listed in Table I will be as listed, the maximum values will not exceed 50 percent of the values listed in Table I.

3.10 Workmanship. The compass shall meet the workmanship requirements of MIL-F-13926. Exterior protective surfaces shall be free of voids which may affect their protective qualities or appearance. There shall be no evidence of flaking, peeling, blisters or holes in the reflecting surface. Blemishes such as discolorations, stains, smears or smudges shall be considered as spots. The maximum apparent dimension of any spot shall not exceed .05 inches. The total number of spots shall not exceed five (5). Spots smaller than .01 inches will be ignored if the number of them in any one inch diameter circle does not exceed five (5) when scanning the full aperture of the mirror. The above sizes, number and distribution of defects shall also apply to the surfaces of the compass box glass cover and the top surface of the mirror. All screw thread fasteners shall be locked in place by lockwashers, plastic pettets (Nylok) or locking compound.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) compasses. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular

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production. All parts and materials, including packaging and packing shall be obtained from the same source of supply as used in regular production. The requirements and tests in Table II shall be met prior to conducting all remaining tests herein.

NOTE: When the first article test is performed, a determination will be made to insure that no magnetic disturbance exists, such as magnetic building material, or test equipment.

TABLE II. First article requirements.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
General description	3.3.1	4.6.1
Configuration	3.3.2	4.6.1
Compass box	3.3.4	4.6.1
Compass needle configuration	3.3.4.4	4.6.1
Scale markings	3.3.4.6	4.6.1

NOTE: When no first article is specified, three (3) compasses shall be selected from the first months production.

4.2.1 Government testing. When the government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection including certificates of conformance for materials, shall be submitted to the government for approval. The government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III, IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to government acceptance or rejection by item or lot. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

TABLE III. Classification of defects.

CLASS	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
CRITICAL:	NONE		
MAJOR:	AQL 1.0% Defective		
101.	Vibration	3.4.4	4.6.5
102.	Reseatability of needle	3.6.1	4.6.8
103.	Sealing	3.4.6	4.6.7
104.	Cleanliness	3.5	Visual
105.	Compass needle horizontal alignment	3.7.1	4.6.17
106.	Compass needle tip clearance	3.7.2	4.6.18
107.	Compass needle tilt clearance	3.6.2	4.6.9
108.	Compass needle damping	3.6.3	4.6.10
109.	Compass needle half-period	3.6.4	4.6.11
110.	Excursion range, declination adjustment	3.6.5.1	4.6.12
111.	Backlash, declination adjustment	3.6.5.2	4.6.12.1
112.	Distortion, mirror	3.6.6	4.6.13
113.	Reflectance	3.6.7	4.6.14
114.	Visibility, scales	3.6.8	4.6.15
115.	Visibility, target	3.6.9	4.6.16
116.	Alignment of sights, azimuth	3.7.3.1	4.6.19
117.	Alignment of sight tips	3.7.3.1.1	4.6.19.1
118.	Alignment of rear sight tip and mirror line	3.7.3.1.2	4.6.19.2
119.	Alignment of sights, elevation	3.7.3.2	4.6.20
120.	Compass level vial setting	3.7.4	4.6.21

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121.	Clinometer level vial setting	3.7.5	4.6.22
122.	Clinometer readings	3.7.6	4.6.23
123.	Azimuth readings	3.7.7	4.6.24
124.	Repeatability of azimuth readings	3.7.8	4.6.25
125.	Operability, cover fastener	3.8.1	4.6.26.1
126.	Torque, rear sight arm hinge	3.8.2	4.6.26.2
127.	Torque, rear sight hinge	3.8.2	4.6.26.2
128.	Torque, front sight hinge	3.8.2	4.6.26.2
129.	Torque, cover hinge	3.8.2	4.6.26.2
130.	Torque, clinometer index and level lever	3.8.2	4.6.26.2
131.	Torque, declination adjusting mechanism	3.8.2	4.6.26.2
132.	Resistance to static charge, transparent compass box cover	3.3.6.3	4.6.1.9
133.	Resistance to static charge, mirror	3.3.6.3	4.6.1.9
134.	Identification	3.3.10	Visual
135.	Workmanship	3.10	Visual-Scale

MINOR: None defined

NOTE: The tests in Table III shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification. Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification for the contractor.

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TABLE III. Certification requirements.

	CHARACTERISTICS	REQUIREMENTS
301.	Materials	3.3.6.1, 3.3.6.2, 3.3.6.3, 3.3.6.4, 3.3.6.5, 3.3.6.6, 3.3.6.7, 3.3.6.8
302.	Final finishes	3.3.7.1, 3.3.7.2, 3.3.7.3, 3.3.7.4
303.	Aluminized surface	3.3.4.1
304.	Protective backing	3.3.4.1
305.	Bonding application	3.3.4.1

4.4 Special sampling.

4.4.1 Construction and environmental. Three compasses shall be selected at random by a Government representative as a special sample from each 50 produced or from each months production, whichever occurs first. The compasses shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table IV when subjected to the temperatures of 3.9.

TABLE IV. Environmental tests.

	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
401.	Cover fastener	3.9	4.6.26.1
402.	Repeatability of azimuth readings	3.7.9	4.6.25.1
403.	Level vials	3.3.8, 3.3.9	4.6.2
404.	Storage temperature range	3.4.1	4.6.3
405.	Condensation	3.4.2	4.6.4
406.	Torque, rear sight arm hinge	3.8.2	4.6.26.2
407.	Torque, rear sight hinge	3.8.2	4.6.26.2
408.	Torque, front sight hinge	3.8.2	4.6.26.2
409.	Torque, cover hinge	3.8.2	4.6.26.2
410.	Torque, clinometer index, level lever	3.8.2	4.6.26.2
411.	Torque, declination adjusting mechanism	3.8.2	4.6.26.2

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4.4.2 Construction and reliability. One compass shall be selected at random by a Government representative as a special sample from each 400 produced, or from each months production, whichever occurs first. The sample shall meet the requirements and tests in Table V.

TABLE V. Construction and reliability tests.

	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
501.	Size and weight	3.3.3	4.6.1.1
502.	Cover and mirror	3.3.4.1	4.6.1.3
503.	Needle lifting mechanism	3.3.4.2	4.6.1.4
504.	Clinometer	3.3.4.5	4.6.1.6
505.	Sight assemblies	3.3.5	4.6.1.8
506.	Shock	3.4.5	4.6.6
507.	Operating life	3.7.9	4.6.25.2

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of inspection" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in

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the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Methods of inspection.

4.6.1 Construction. The compass shall be visually examined for conformance with the general description and configuration requirements of Table II. For standard sampling testing, a visual and tactile comparison shall be made with approved preproduction models to determine if unauthorized changes in materials, basic construction or functional characteristics have been made.

4.6.1.1 Size and weight. The dimensional characteristics on Figure 1 and 2 shall be measured with standard measuring equipment or gages. Total weight shall be measured on a scale having an accuracy within one ounce to determine compliance with 3.3.3.

4.6.1.2 Compass box. The compass box cover shall be visually examined for flaws, bubbles, scratches, haze or discoloration. Any evidence of these defects shall be considered as failing to meet the transparency requirements of 3.3.4. All scales, indexes, and level bubbles shall be examined for visibility and legibility using a fluorescent desk lamp, at least 30 watts, set one foot above the compass.

4.6.1.3 Cover and mirror. Installation of mirror shall be checked for compliance with 3.3.4.1 by checking alignment of the rear sight tip with each end of the etched line. Firmness shall be checked by applying a torsional pressure with the thumbs. This pressure shall be applied in both clockwise and counterclockwise directions parallel to the top surface of the mirror in an effort to rotate the mirror. The mirror shall not move. Any movement of the mirror will be detected when the rear sight tip fails to coincide with the etched line. The dimensional characteristics for the compass cover transparent area shown on Figure 1 shall be measured using standard measuring equipment. The transparent area shall be visually examined for flaws, bubbles, scratches, haze or discoloration under a fluorescent lamp of at least 30 watts. Any evidence of these defects, which would interfere with sighting or affect image quality, shall be cause for rejection and suspension of further acceptance. When bonding is used, visually inspect for application of cement between top surface of mirror and cover.

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4.6.1.4 Needle lifting mechanism. With the cover closed, shake the compass gently in a plane perpendicular to the face of cover glass and listen for any rattle to assure that the compass needle is held in compliance with the requirements of 3.3.4.2.

4.6.1.5 Cover fastener. Proper latching and unlatching of the cover in the closed position shall be determined utilizing a spring pull test scale to determine torque or by a standard type pressure gage to determine pressure, as indicated in 3.3.4.3, whichever is applicable in testing for compliance with 3.3.4.3.

4.6.1.6 Clinometer. Compliance with the requirements of 3.3.4.5 shall be determined by a visual and manual examination. The clinometer index mechanism shall be actuated by manual means so that the index moves to both extremes on the angle of site scale. During this operation, there shall be no excessive binding or looseness. It shall also be noted, that the mechanism shall have remained in any set position after being shaken in a plane parallel to the face of the window.

4.6.1.7 Scales. Azimuth and elevation scales shall be visually examined for quality and legibility and for conformance with Figures 3a and 3b. The external azimuth scale shall be actuated to determine compliance with the requirements of 3.3.4.6.

4.6.1.8 Sight assemblies. Conformance with the requirements of 3.3.5 for rear sight arm, rear sight, front sight, and clearance between sights and glass shall be determined by a visual and manual examination. The hinged components shall be actuated by manual means. During this operation, movement and any excessive binding or looseness will be noted. Mechanical indicators may be used to determine that the hinged components can be opened and set within the tolerance specified in Figure 2.

4.6.1.9 Resistance to static charge. The test shall be performed utilizing a horizontal surface, a clean dry cotton cloth and a standard time piece. The compass in the open position shall be held on the horizontal surface. The north seeking tip of the compass needle shall indicate 3200 mils on the azimuth scale, prior to subjecting the compass to the following, to determine compliance with 3.3.6.3.

4.6.1.9.1 Mirror. The surface of the mirror shall be rubbed briskly with the cloth for a minimum of 25 strokes in a circular motion, followed by manually setting the hinged cover within 30 degrees from the closed position as measured with the aid of a protractor. Immediately after rubbing, rotate compass by 1600 $\pm$ 10 mils either clockwise or counterclockwise making sure the needle is not jarred or vibrated. The north seeking tip shall indicate either 4800 $\pm$ 20 mils or 1600 $\pm$ 20 mils within 15 seconds after rubbing operation is completed.

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4.6.1.9.2 Transparent compass box cover. With the compass in the open position, the surface of the transparent compass box cover shall be rubbed briskly with the cloth for a minimum of 25 strokes in a circular motion. Immediately after rubbing, rotate compass by 1600 ± 10 mils either clockwise or counterclockwise making sure the needle is not jarred or vibrated. The north seeking tip shall indicate either 4800 ± 20 mils or 1600 ± 20 mils within 15 seconds after rubbing operation is completed.

4.6.2 Compass level vials. Sensitivity of the level vials shall be inspected as required by MIL-V-3144 at the operating temperatures of 3.9.

4.6.3 Storage temperature range. The compass shall be exposed to the temperature conditions specified in 3.4.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. After the compass has been exposed to the extreme temperatures and permitted to return to standard ambient temperature, the compass shall be examined for evidence of physical failure such as specified in 3.4.1. This test can be performed in conjunction with 4.6.2.

4.6.4 Condensation. The compass shall be placed within a test chamber and the chamber cooled to and maintained at the temperature specified in 3.4.2 until stabilization is reached. While at this temperature, it shall be determined by a visual examination with the unaided eye, that the interior of the compass box conforms with the requirements of 3.4.2.

4.6.5 Vibration. Vibration tests shall be made on a type of vibrating machine capable of meeting the conditions of 3.4.4. The accuracy of the applicable test equipment and test equipment measurements shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The compass shall be secured to the vibrating table in the position specified in 3.4.4, leaving the compass cover free from any holding media. The compass shall be vibrated successively along each of the three mutually perpendicular directions that are respectively parallel to the edges of the compass with the motion and frequency conforming to the specified condition in 3.4.4 applied for a period of 5 minutes ± 15 seconds.

Subsequent to vibration, the compass assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.4 and 3.8.1, prior to being subjected to the remaining test specified herein. A visual examination shall be made for missing or damaged parts, glass chips, foreign matter or volatile substances.

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A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts resulting from the test. Anyone of the aforementioned defects shall be cause to consider the compass defective. Compasses that have passed the above examinations shall then be subjected to the remaining test herein.

4.6.6 Shock. The compass in the closed position shall be subjected to two 36 inch drops onto a solid wood floor covered with 4 inches of 40 grit kiln-dried sand. One of the two drops shall be with impact on the bottom of the compass box. The other impact on any of the four sides of the compass box. After each drop the compass shall be examined by visual and tactile means to ascertain that there is no evidence of physical defects as outlined in 4.6.5, and all manually operated parts can be operated without undue force and will remain in the desired position to meet the requirements of 3.4.5.

4.6.7 Sealing. The compass shall show no evidence of leakage when submitted to the requirements of 3.4.6.

4.6.8 Reseatability of needle. To determine compliance with the requirements of 3.6.1, the compass shall be inverted to its normal upright position. The needle shall reset itself and swing freely on its pivot without tapping, shaking, or any advertent movement of the compass.

4.6.9 Compass needle tilt clearance. The opened compass shall be positioned on a holding media oriented on a N-S axis at a 3 degree angle, with the north seeking tip indicating zero. The needle shall be magnetically deflected off the zero position a minimum of 200 mils. Upon removal of the magnetic influence, the needle shall return to a zero position within 10 mils. Repeat the test with the compass at a 3 degree angle in the "E-W", "S-N", and "W-E" quadrants.

4.6.10 Compass needle damping. The open compass shall be positioned on a leveled surface plate with the north seeking tip indicating at zero on the azimuth scale. The compass needle shall be magnetically deflected 400 mils from the zero position, and released. The time required for the needle to come to rest shall be determined by means of a commercial stop watch. The time noted shall be within the 15 seconds specified in 3.6.3.

4.6.11 Compass needle half-period. Using the same set-up as in 4.6.10, magnetically deflect needle a maximum of 400 mils. The average half-period time shall be determined by means of a commercial stop watch. Should the average exceed the requirements of 3.6.4, an investigation shall be conducted to determine the presence of external magnetic influence or possibility of a weak natural magnetic field.

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4.6.12 Declination adjustment, excursion range. Utilizing a screw driver, rotate the declination adjusting screw through its maximum limits to determine compliance with the excursion requirements of 3.6.5.1.

4.6.12.1 Backlash. Backlash shall be determined by rotating the adjusting screw in one direction and recording a selected setting. Mark the positive setting of the screw head with relation to the compass box and then continue rotating the screw in the same direction for a minimum of 200 mils. Reverse screw rotation and without changing direction return the screw to the exact marked position (with proper rotations from originally marked position). The pre-recorded selected reading on the azimuth scale shall be observed within the 20 mils specified.

4.6.13 Distortion, mirror. With the compass placed on a surface three feet from the viewer, and the cover adjusted so that the entire face of the compass is visible, there shall be no visible evidence of distortion, and all lines, numerals, and letters shall be discernable to meet the requirements of 3.6.6.

4.6.14 Reflectance. The test shall be conducted in accordance with the reflectivity test procedure outlined in MIL-M-13508 utilizing the equipment or equal as illustrated diagrammatically on Figure 1 attached. The mirror shall reflect no less than the minimum percentage specified in 3.6.7.

4.6.15 Visibility, scales. The compass shall be visually examined for compliance with the requirements of 3.6.8. Sight through the transparent cover, with the bottom of the open compass on a horizontal plane; and then with the "W" side of the compass box and cover on a horizontal plane. Repeat the examination utilizing a light source specified for the test in 4.6.1.2.

4.6.16 Visibility, target. Test equipment required to conduct the test shall consist of an appropriate resolution target. The target shall be viewed from a distance of 10 feet $\pm$ 1 foot. Sight through the transparent area and observe the image of the target to determine compliance with the requirements of 3.6.9.

4.6.17 Compass needle horizontal alignment. Position the compass on a level nonmagnetic surface so that the north seeking tip indicates "zero" on the azimuth scale. The north seeking tip will be examined with the unaided eye to determine if the tip is in the same horizontal plane as the top of the azimuth ring. If the needle tip is elevated or depressed it will be placed on an angle block that will displace the tip in the direction of the top of azimuth ring by .030 of an inch, the needle tip will be examined to determine if the tip has passed through or is in the same horizontal plane as the top of the azimuth ring. If the tip fails to pass through or to be in the same horizontal plane as the top of the azimuth ring it will be considered as a reject.

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The test will be repeated for the south seeking tip at 3200 mils. Repeat the entire procedure, with the north seeking tip indicating 1600, 3200, and 4800 mils on the azimuth scale to determine compliance with the requirements of 3.7.1.

4.6.18 Compass needle tip clearance. The opened compass shall be placed on a leveled non-magnetic surface plate. A microscope with a reticle providing a minimum magnification of 8 power will be utilized to determine compliance with the requirements of 3.7.2. The reticle pattern shall be calibrated to permit measurement for the clearance requirement of 3.7.2. This requirement shall be met with the north seeking tip indicating 0, 1600, 3200, and 4800 mils on the azimuth scale.

4.6.19 Alignment of sights, azimuth. The compass opened as shown in Figure 2 (except that tips should be opened to 180°), shall be positioned with the bottom of the compass on a leveled surface plate. The test is accomplished with the aid of a transparent gage with two parallel lines scribed at the tolerance specified in 3.7.3.1. The sight tips, azimuth scale index, azimuth zero and 3200 mil graduations and etched line of the mirror should fall between the two parallel lines.

4.6.19.1 Alignment, sight tips. The compass opened as shown in Figure 2, shall be positioned with the bottom of the compass on a level surface plate. The test is accomplished with the aid of a vertical adjustable microscope containing a reticle pattern with tolerance lines to permit measurement for the alignment requirement, a straight edge and a vernier height gage. Move front and rear sights so that the extreme points of the sighting tips make contact with each other. Sight through the viewing microscope and adjust for best focus of the tips. When this is attained, the sharp point of the tips shall be aligned within the limit specified in 3.7.3.1.1 as measured on the microscope reticle.

4.6.19.2 Alignment, rear sight tip and mirror line. The test shall be performed with the sight tips fully extended with the compass secured, and positioned as specified in 4.6.19.1. Position the viewing microscope over the sight tip of the compass cover. Sight through the microscope and observe the sharp point of the tip, it shall be aligned with the intersecting point of the crosslines of the microscope reticle. Reposition the viewing microscope over the rear sight and observe the sharp point of the sight tip, it shall be aligned with the intersecting point of the crosslines of the microscope reticle within the limit specified. To check the parallel travel or tracking ability of the rear sight, move the sighting arm in increments of approximately 30° through its full excursion. At each point, reposition the microscope and readjust the focus at each point. The sharp point of the tip shall track the intersecting point of the reticle crosslines within the limit specified. Repeat this procedure for

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the front sight tip, and it shall also track within the limit specified 3.7.3.1.2 as measured on the microscope reticle. Reposition the sight tips as shown in Figure 2. The compass shall then be subjected to the test in 4.6.20.

4.6.20 Alignment of sights elevation. Subsequent to the test in 4.6.19.2, position the "W" side of the compass box and cover on the surface plate. The distance from each of the sight tips to the surface plate shall be measured by means of the vernier height gage and the difference in measurement shall be within the tolerance specified in 3.7.3.2.

4.6.21 Compass level vial setting. The leveling requirement shall be determined by a visual and manual examination. The open compass shall be positioned with the bottom of the compass box on a leveled surface plate, with the clinometer index set at "zero". The bubble in the circular level vial shall be centered to conform with the requirement of 3.7.4. Should the circular level vial be mounted on a moving part, the bubble shall coincide with the red circle within the limits specified in 3.7.4 when the part is positioned at each limit of travel.

4.6.22 Clinometer level vial setting. Place the open compass with the side of the compass as indicated in 3.7.5 down on a leveled surface plate. Adjust clinometer level to center the bubble in the vial. The clinometer index shall coincide with the zero graduation within the limit specified in 3.7.5.

4.6.23 Clinometer readings. With the compass opened as shown in Figure 2, position the "W" side of compass box and cover on a calibrated fixture. The fixture shall be adjustable through a range of 1200 mils in elevation and depression. Readings shall be taken at 400, 800, and 1200 mils in elevation and depression. The clinometer shall indicate the settings within the limits specified in 3.7.6.

4.6.24 Azimuth readings. The test shall be accomplished using an indexing table accurate to at least one mil, or wall targets established by a surveyor's transit. Means shall be provided to insure that the compass is rotated about the needle point. Measurements shall be taken at three azimuth positions approximately equally spaced in 6400 mils. The readings shall be within the limits specified in 3.7.7 for the north and south seeking tips.

4.6.25 Repeatability of azimuth readings (60° to 90°F). The test shall be performed in conjunction with the test in 4.6.24. At each azimuth position established in 4.6.24, the compass needle shall be magnetically deflected 500 + 300 mils and released. The compass needle shall be observed to determine compliance with the requirements of 3.7.8 for the standard temperature condition.

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4.6.25.1 Repeatability of azimuth readings (-40° and +150°F). The compasses used for the test in 4.6.2 shall be exposed to the extreme temperatures specified in 3.7.8, employing the test facilities used for the temperature test of 4.6.3. The temperature shall be changed gradually to avoid thermal shock. When thermal stabilization is reached at each temperature extreme, the compass shall then be tested in accordance with the procedure employed in 4.6.25 to determine compliance with the requirements for the extreme temperature condition.

4.6.25.2 Operating life. The compass needle shall be raised and lowered by means of the needle lifting mechanism. Repeat this complete cycle 15,000 times. After each 1000 cycles the test shall be stopped and the compass shall be visually examined to determine that the needle is firmly locked against the window when the plunger is depressed, and that the needle swings freely on its pivot when the plunger is released. Any failure of the needle lifting mechanism during this test shall be cause for rejection.

4.6.26 Operability of moving parts (60° to 90°F).

4.6.26.1 Effectiveness of cover fastener. Unlatching of the cover in the closed position shall be determined utilizing a commercial type direct force, push or pull measuring device graduated in increments of ounce-inches, whichever is applicable in testing for compliance with 3.8.1. The amount of force shall be recorded.

4.6.26.2 Hinges and adjustments. The maximum torque required to operate the manually adjustable components shall be measured using a commercial type direct force, push or pull, spring scale device graduated in increments of ounce-inches. The device shall be equipped with suitable adapters to accomodate testing of the specified components, as applicable. The torque required to operate the declination adjustment screw shall be measured using a torque-indicating wrench type tester graduated in ounce-inches, and equipped with an adapter to accomodate testing the clinometer index and level lever.

The minimum cover hinge torque shall be met if the hinge is capable of supporting the weight of the body when the cover is open to a 120 degree (approximately) angle (with respect to the cover glass) and the cover is held horizontal.

The minimum rear sight arm hinge torque shall be met if the hinge is capable of supporting the weight of the body when the cover is open to a maximum angle of 15 degrees (approximately) (with respect to the cover glass) and the rear sight arm is open to a 135 degree (approximately) angle and held horizontal.

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The minimum torque for the rear sight hinge, front sight hinge, clinometer index and declination adjusting mechanism shall be met if the mechanism remains in any set position after being shaken gently through an arc of 120 degrees (approximately) in a plane parallel to the face of the window. The compass shall not be impacted during this test. The torques shall be within the values listed in Table I of 3.8.2 while at standard temperature, and the operability temperatures of 3.9.

4.6.26.3 Operability of moveable parts (-40° and +150°F). During the performance testing at the temperatures specified in 3.9, the torque of the specified components shall be measured and shall not vary from the recorded torque values in 4.6.26.1 and 4.6.26.2 by more than the tolerance specified in 3.9. The test shall be conducted as specified in 4.6.26.1 and 4.6.26.2.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with packaging data sheet 10547166. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. The Compass, Magnetic, Unmounted: M2 is a multipurpose instrument consisting of a clinometer and magnetic compass. The clinometer provides for measuring vertical angles and angles of site. The compass provides for measuring grid and magnetic azimuths, and horizontal angles. This instrument may also be utilized to check flat surfaces for level within the accuracy limits provided by the circular level.

6.2 Ordering data. Procurement documents should specify:

a. Title, date, and number of this specification.

b. Serial numbers to be used for identification marking. (See 3.2.10.3)

c. Applicable packaging data sheet number.

d. Selection of applicable levels of preservation, and packaging.

e. Requirements for first article testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-10054.

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6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001. When the contractor submits inspection equipment designs to the government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.)

c. The designs remaining to be submitted and the expected date of submittal.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project No. 1290-A356)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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